

on the classic ESP32 (real time)

core 0 : acquisition

core 1 : inference

2.4 GHz AP

(router / phone)

ping, ~ 100 pkt/s

**Wi-Fi RX +
CSI callback**

ring buffer

52 subcarrier
amplitudes

window $T=64$

z-score +
int8 quantize

TinyCNN-16

TFLite Micro

**activity
label**

Measured, 30 min continuous: 180,815 CSI packets (~ 100 pkt/s) · 2,825 windows · 0.00%
loss · 37.6 ms end-to-end / window · 1.57 windows/s · heap stable (± 2.5 kB) · 0 crashes